

Intro and k -means clustering

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Branches of Machine Learning

- ▶ **Supervised learning:**
 - ▶ **Classification** - how to predict a categorical outcome variable
 - ▶ **Regression** - how to predict a numerical outcome variable
- ▶ **Unsupervised learning:**
 - ▶ **Clustering** - how to categorize the *observations* into meaningful groups (that aren't predefined)
 - ▶ **Dimension reduction** - how to meaningfully represent each observation using $p^* < p$ features by finding patterns among the original features
 - ▶ **Anomaly detection** - how to identify observations that deviate significantly from “normal data”

First steps

- ▶ We'll devote most of our attention this semester to supervised learning, but we'll start with a short unit on unsupervised learning for a few reasons:
 - ▶ It's manageable to learn: Python, unsupervised learning algorithms, and notation simultaneously
 - ▶ Unsupervised methods can be used as *pre-processing steps* to prepare data for supervised learning

Notation

To begin, let's consider **tabular data** that is organized in a matrix, **X**, consisting of n observations (rows) with p features (columns)

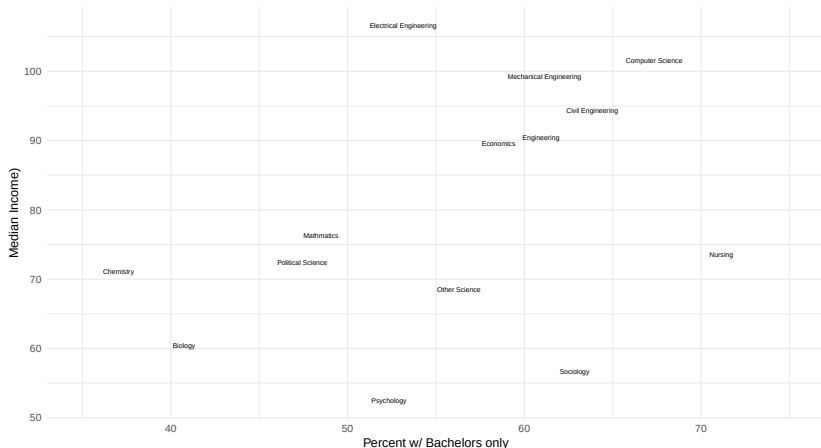
$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,p} \end{pmatrix}$$

In this class I'll try to stick to the following conventions:

1. Bold capital letters to denote a matrix
2. Bold lower-case letters to denote a vector
3. non-bold capital letters to denote a random variable
4. non-bold lower-case letters to denote a scalar value

Example Data

These are data from Census's ACS income by degree field survey.
Note this data may be out of date.



Clustering

- ▶ The goal of clustering is categorize our n observations into groups
- ▶ The first approach we'll discuss is **k-means clustering**: find k non-overlapping groups, where k is **pre-specified**

k-means: Pseudocode

1. Pre-specify k : the number of clusters
2. Randomly assign k initial cluster centers
3. Assign each observation to the closest cluster center
4. Re-calculate the k cluster centers

Repeat previous two steps until cluster assignments “stop changing”
i.e. algorithm “has converged”

Objective function

What do we mean by “stop changing” and “has converged”

- ▶ In general: machine learning methods occur by optimizing an **objective function** also called cost and loss functions
- ▶ Ex: Linear regression seeks to minimize the sum of squared residuals

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ▶ *k*-means clustering seeks to minimize the total within cluster sum of squared euclidean distance between observations and their cluster center

k-means: Visual demo

- ▶ Here is a [k-means GIF animation](#) (rather fast)
- ▶ Let's break this down frame by frame. Go to `demo/frames` and open all images at once

k-means: More formally

- ▶ Consider a set of **prototypes**, i.e. the cluster centers, defined by the following coordinates in the p -dimensional space of our data: $\{\mathbf{z}_1, \dots, \mathbf{z}_k\}$
- ▶ Let $A()$ denote an *assignment function* that assigns observations to the group associated with a prototype i.e. $A()$ assigns observations to clusters
- ▶ For example, $A(\mathbf{x}_i) = j$ indicates the i^{th} observation is assigned to the j^{th} prototype

k-means: Objective function

- ▶ In *k*-means clustering, we seek to minimize the following:

$$\sum_{i=1}^n \|\mathbf{x}_i - \mathbf{z}_{A(\mathbf{x}_i)}\|^2$$

- ▶ Or using linear algebra notation

$$\sum_{i=1}^n (\mathbf{x}_i - \mathbf{z}_{A(\mathbf{x}_i)})^T (\mathbf{x}_i - \mathbf{z}_{A(\mathbf{x}_i)})$$

- ▶ The data (the $\mathbf{x}_i$'s) are fixed, and we can simplify things by assuming an assignment function, such as the nearest prototype by euclidean distance: $d(\mathbf{x}_1, \mathbf{x}_2) = \sqrt{\sum_{j=1}^k (x_{1,j} - x_{2,j})^2}$
- ▶ *k*-means clustering amounts to solving for the prototype coordinates that minimize this objective function

k-means: More formal pseudocode

The *k*-means algorithm alternates between two steps:

1. Given the current prototype coordinates, assign each observation to the nearest prototype using euclidean distance (ie: apply $A()$ to each observation).
2. Update the prototype coordinates to now be the mean coordinates of their respective assigned observations.

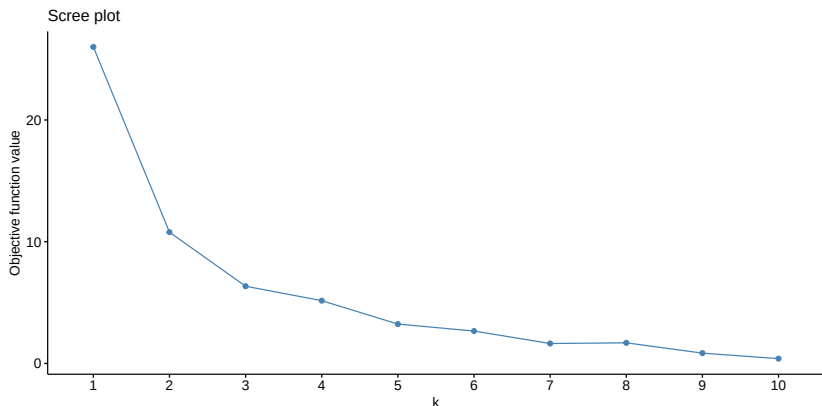
These steps repeat until additional iterations no longer improve the objective function.

k-means: Role of *k*

- ▶ The *k*-means algorithm requires a pre-specified value of a *k*, making it an example of what's known as a **hyperparameter** or **tuning parameter**
- ▶ A simple way to intelligently decide upon the value of hyperparameter is a method known as **grid search**, which is an exhaustive search over an enumerated set of candidate values
- ▶ In the context of *k*-means, we might look at the value of the objective function over a sequence of *k* values
- ▶ We can then visually display this relationship in a scree plot and select *k* at an inflection point (where additional prototypes no longer produce the same improvement in the object function)

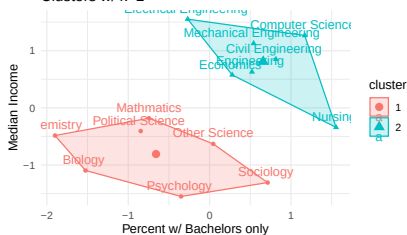
Example Data: Scree plot for example data

Below is the value of the objective function for several choices of k for our example data, notice how it levels off:

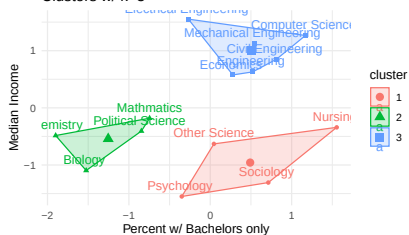


Example Data: Results for Different k

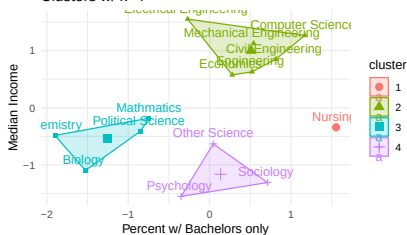
Clusters w/ $k=2$



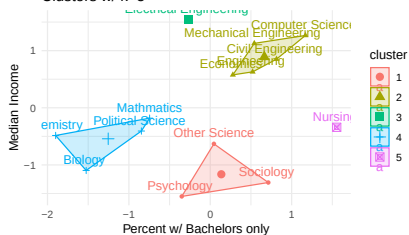
Clusters w/ $k=3$



Clusters w/ $k=4$



Clusters w/ $k=5$



k-means: Python implementation

Unfortunately, *k*-means is sensitive to the initial set of prototype coordinates. The `sklearn` implementation allows us two options:

1. `random` (also known as Forgy) - randomly select k of the observed observations and use their coordinates as the initial prototypes
2. `k-means++` - choose the first prototype uniformly at random from the observed observations, then use probability distributions find additional prototypes that are far from existing prototypes ([click here for details](#))

Conclusion

- ▶ *Supervised learning* aims to predict a numeric outcome or classify data-points after learning the patterns present in the available data. i.e. there is a y outcome variable
- ▶ *Unsupervised learning* finds patterns in the available data without specifying a target outcome i.e. there is no outcome variable. k -means clustering is an example of an unsupervised learning method